QUERY PERFORMANCE PROJECT

Performance Improvement & Recommendation Validation

TC-01 to TC-25 Complete Status Report with Proofs

Final Result	**25 PASS / 0 FAIL**
Pass Rate	**100%**
Primary Endpoint	POST /regression-check
Supporting Endpoints	POST /regression-history-check; POST /multiple-regression-check
Database	SQLite

FINAL STATUS: 25 / 25 PASS

Evidence basis: textual Swagger JSON response payloads supplied during the interactive test run. HTTP 200 alone was not used as PASS evidence.

1. Executive Summary

All 25 screenshot-defined Performance Improvement / Recommendation Validation test cases reached final PASS status. The suite validates runtime improvement detection, improvement thresholds, plan-level evidence, recommendation success/failure, workload-wide improvement, planning-time separation, row-processing evidence, before/after comparison, and rollback guidance.

Final status reflects the latest successful retest for cases that initially failed. Minimum targeted backend changes were applied and each affected case was retested before being counted as PASS.

2. Final Status Matrix

ID	Test case	Final proof (key evidence)	Status
TC-01	Query becomes faster	old_execution_time = 5000; new_execution_time = 1000	**PASS**
TC-02	Index improves query	old_scan_method = "Seq Scan"; new_scan_method = "Index Scan"	**PASS**
TC-03	No improvement	old_execution_time = 5000; new_execution_time = 4900	**PASS**
TC-04	Query becomes slower	old_execution_time = 1000; new_execution_time = 3000	**PASS**
TC-05	Large improvement	old_execution_time = 10000; new_execution_time = 1000	**PASS**
TC-06	Small improvement	old_execution_time = 5000; new_execution_time = 4800	**PASS**
TC-07	Execution plan improves	plan_changed = true; old_scan_method = "Seq Scan"	**PASS**
TC-08	Plan changes but time does not	plan_changed = true; scan_method_changed = true	**PASS**
TC-09	Time improves but plan stays the same	plan_changed = false; old_execution_time = 3000	**PASS**
TC-10	Multiple executions	old_average = 1000; new_average = 500	**PASS**
TC-11	Index recommendation validation	old_execution_time = 4000; new_execution_time = 1200	**PASS**
TC-12	Composite index validation	new_query_plan uses "idx_orders_customer_status"; scan_method_changed = true	**PASS**
TC-13	Wrong recommendation	improvement_percent = 2; improvement_level = "insignificant"	**PASS**
TC-14	Multiple tuning actions	regression_cause includes Query planner configuration changed; regression_cause includes Statistics refreshed / ANALYZE event	**PASS**
TC-15	Improvement exceeds threshold	old_execution_time = 5000; new_execution_time = 1000	**PASS**
TC-16	Improvement below threshold	old_execution_time = 5000; new_execution_time = 4800	**PASS**
TC-17	Query already fast	plan_changed = false; improvement_percent = 2	**PASS**
TC-18	Improvement disappears later	old_average = 1000; new_average = 1764	**PASS**

Performance Improvement / Recommendation Validation - TC-01 to TC-25

ID	Test case	Final proof (key evidence)	Status
TC-19	Improvement for one query	query_name = "TC19_ORDERS_QUERY"; improvement_percent = 75	**PASS**
TC-20	Workload-wide improvement	Q_A performance_ratio = 0.25; Q_B performance_ratio = 0.4	**PASS**
TC-21	Planning time improvement	old_planning_time = 500; new_planning_time = 100	**PASS**
TC-22	Execution time improvement	old_execution_time = 3000; new_execution_time = 1500	**PASS**
TC-23	Rows processed decrease	old_actual_rows = 10000; new_actual_rows = 1000	**PASS**
TC-24	Before/after report	old/new query plans returned; cost_difference = -480	**PASS**
TC-25	Recommendation rollback	improvement_percent = 1; improvement_level = "insignificant"	**PASS**

3. Detailed Proof Evidence

Each case below lists the response fields that were used for the final PASS verdict. Proof is limited to the Swagger payloads supplied in chat; no independent database execution is asserted beyond those responses.

TC-01 - Query becomes faster

Status	**PASS**
Expected	Detect improvement

Observed proof

- old_execution_time = 5000
- new_execution_time = 1000
- performance_ratio = 0.2
- improvement_detected = true
- regression_detected = false

Assessment: Runtime decreased by 80%; the endpoint correctly classified the change as an improvement.

TC-02 - Index improves query

Status	**PASS**
Expected	Report improvement after index use

Observed proof

- old_scan_method = "Seq Scan"
- new_scan_method = "Index Scan"
- plan_changed = true
- added_plan_nodes includes "idx_orders_customer_id"
- performance_ratio = 0.2
- improvement_detected = true

Assessment: The new index appeared in the plan and coincided with a large runtime improvement.

TC-03 - No improvement

Status	**PASS**
Expected	Do not report a tiny change as significant

Observed proof

- old_execution_time = 5000
- new_execution_time = 4900
- performance_ratio = 0.98
- improvement_percent = 2
- improvement_detected = false

Assessment: After adding a configurable improvement threshold, a 2% change was correctly treated as insignificant.

TC-04 - Query becomes slower

Status	**PASS**
Expected	Report regression

Observed proof

- old_execution_time = 1000
- new_execution_time = 3000
- performance_ratio = 3
- regression_detected = true
- result = "Regression detected"

Assessment: The 3x slowdown exceeded the regression threshold and was detected.

TC-05 - Large improvement

Status	**PASS**
Expected	Report high improvement

Observed proof

- old_execution_time = 10000
- new_execution_time = 1000
- improvement_percent = 90
- improvement_level = "high"
- improvement_detected = true

Assessment: The endpoint explicitly classified the 90% gain as a high improvement.

TC-06 - Small improvement

Status	**PASS**
Expected	Apply configured improvement threshold

Observed proof

- old_execution_time = 5000
- new_execution_time = 4800
- improvement_percent = 4
- improvement_level = "insignificant"
- improvement_detected = false

Assessment: A 4% gain did not cross the configured 10% threshold.

TC-07 - Execution plan improves

Status	**PASS**
Expected	Record plan improvement

Observed proof

- plan_changed = true
- old_scan_method = "Seq Scan"
- new_scan_method = "Index Scan"
- removed_plan_nodes includes "Seq Scan"
- improvement_percent = 66.67
- improvement_detected = true

Assessment: Plan transition and runtime gain were both captured.

TC-08 - Plan changes but time does not

Status	**PASS**
Expected	Do not claim meaningful improvement

Observed proof

- plan_changed = true
- scan_method_changed = true
- old_execution_time = 1000
- new_execution_time = 1000
- improvement_percent = 0
- improvement_detected = false

Assessment: A plan change alone did not produce a false improvement finding.

TC-09 - Time improves but plan stays the same

Status	**PASS**
Expected	Report runtime improvement

Observed proof

- plan_changed = false
- old_execution_time = 3000
- new_execution_time = 1500
- improvement_percent = 50
- improvement_level = "high"
- improvement_detected = true

Assessment: The system recognized runtime improvement even without a plan change.

TC-10 - Multiple executions

Status	**PASS**
Expected	Compare stable aggregated performance

Observed proof

- old_average = 1000
- new_average = 500
- performance_ratio = 0.5
- regression_detected = false
- result = "No regression"

Assessment: Multiple execution samples were aggregated rather than relying on one run.

TC-11 - Index recommendation validation

Status	**PASS**
Expected	Measure before/after impact

Observed proof

- old_execution_time = 4000
- new_execution_time = 1200
- plan_changed = true
- scan_method_changed = true
- improvement_percent = 70
- recommendation_status = "successful"

Assessment: The recommendation was validated with both plan and runtime evidence.

TC-12 - Composite index validation

Status	**PASS**
Expected	Measure composite-index improvement

Observed proof

- new_query_plan uses "idx_orders_customer_status"
- scan_method_changed = true
- improvement_percent = 70
- improvement_level = "high"
- improvement_detected = true

Assessment: The composite index was visible in the plan and delivered a high improvement.

TC-13 - Wrong recommendation

Status	PASS
Expected	Report unsuccessful recommendation

Observed proof

- improvement_percent = 2
- improvement_level = "insignificant"
- improvement_detected = false
- recommendation_status = "unsuccessful"

Assessment: After adding explicit recommendation status, a non-beneficial recommendation was marked unsuccessful.

TC-14 - Multiple tuning actions

Status	PASS
Expected	Compare results carefully when several actions occur

Observed proof

- regression_cause includes Query planner configuration changed
- regression_cause includes Statistics refreshed / ANALYZE event
- regression_cause includes Index created during observation
- improvement_percent = 76
- recommendation_status = "successful"

Assessment: The response preserved multiple tuning signals and still quantified the observed improvement.

TC-15 - Improvement exceeds threshold

Status	PASS
Expected	Mark recommendation successful

Observed proof

- old_execution_time = 5000
- new_execution_time = 1000
- improvement_percent = 80
- improvement_level = "high"
- recommendation_status = "successful"
- improvement_detected = true

Assessment: The improvement comfortably exceeded the configured threshold and the recommendation was marked successful.

TC-16 - Improvement below threshold

Status	PASS
Expected	Mark as insignificant

Observed proof

- old_execution_time = 5000
- new_execution_time = 4800
- improvement_percent = 4
- improvement_level = "insignificant"
- improvement_detected = false

Assessment: The improvement remained below the configured threshold.

TC-17 - Query already fast

Status	PASS
Expected	Avoid unnecessary recommendation

Observed proof

- plan_changed = false
- improvement_percent = 2
- improvement_level = "insignificant"
- recommendation_status = "not_applicable"
- regression_detected = false

Assessment: No tuning recommendation was treated as applicable for an already-fast stable query.

TC-18 - Improvement disappears later

Status	PASS
Expected	Detect later regression

Observed proof

- old_average = 1000
- new_average = 1764
- performance_ratio = 2.5
- regression_detected = true
- result = "Regression detected"

Assessment: Later execution samples showed the earlier gain did not persist; the regression was detected.

TC-19 - Improvement for one query

Status	**PASS**
Expected	Record query-specific result

Observed proof

- query_name = "TC19_ORDERS_QUERY"
- improvement_percent = 75
- improvement_level = "high"
- recommendation_status = "successful"
- improvement_detected = true

Assessment: The result remained attached to the specific query identity.

TC-20 - Workload-wide improvement

Status	**PASS**
Expected	Report broader improvement across queries

Observed proof

- Q_A performance_ratio = 0.25
- Q_B performance_ratio = 0.4
- Q_C performance_ratio = 0.3
- workload_wide_improvement = true
- workload_improvement_reason = "Multiple queries improved together - broader workload improvement"

Assessment: After adding workload aggregation, multiple simultaneous query gains were explicitly reported as a broader improvement.

TC-21 - Planning time improvement

Status	**PASS**
Expected	Record planning improvement separately

Observed proof

- old_planning_time = 500
- new_planning_time = 100
- planning_improvement_percent = 80
- planning_improvement_detected = true
- execution improvement_detected = false

Assessment: Planning-time improvement was separated from execution-time improvement.

TC-22 - Execution time improvement

Status	**PASS**
Expected	Record execution improvement clearly

Observed proof

- old_execution_time = 3000
- new_execution_time = 1500
- improvement_percent = 50
- improvement_detected = true
- planning_improvement_detected = false

Assessment: Execution improvement was isolated from unchanged planning time.

TC-23 - Rows processed decrease

Status	**PASS**
Expected	Report row reduction as supporting evidence

Observed proof

- old_actual_rows = 10000
- new_actual_rows = 1000
- actual_rows_difference = -9000
- scan_method_changed = true
- improvement_percent = 70
- improvement_detected = true

Assessment: Row-processing reduction provided supporting evidence for the measured performance gain.

TC-24 - Before/after report

Status	**PASS**
Expected	Generate a complete comparison report

Observed proof

- old/new query plans returned
- cost_difference = -480
- actual_rows_difference = -9000
- old_execution_time = 5000
- new_execution_time = 1200
- improvement_percent = 76
- planning_improvement_percent = 40

Assessment: The response combined plan, cost, rows, execution, planning, and improvement evidence in one before/after result.

TC-25 - Recommendation rollback

Status	PASS
Expected	Suggest rollback or further investigation

Observed proof

- improvement_percent = 1
- improvement_level = "insignificant"
- recommendation_status = "unsuccessful"
- recommendation_action = "Rollback recommendation or investigate further"
- improvement_detected = false

Assessment: An unsuccessful recommendation produced an explicit rollback/further-investigation action.

4. Corrective Retest Ledger

The following cases initially failed their screenshot-defined expectation, received the minimum targeted change, and then passed on retest.

Case	Minimum fix	Retest proof	Final
TC-03	Added improvement_threshold (default 10%) and changed improvement detection to require the threshold.	Retest: improvement_percent = 2; improvement_detected = false.	**PASS**
TC-05	Added improvement_percent and improvement_level fields.	Retest: improvement_percent = 90; improvement_level = "high".	**PASS**
TC-13	Added recommendation_status classification.	Retest: recommendation_status = "unsuccessful".	**PASS**
TC-20	Added workload_wide_improvement and workload_improvement_reason to /multiple-regression-check.	Retest: workload_wide_improvement = true with explicit broader-workload reason.	**PASS**
TC-21	Added separate planning-time input, calculation, and output fields.	Retest: planning_improvement_percent = 80; planning_improvement_detected = true.	**PASS**
TC-25	Added recommendation_action for unsuccessful recommendations.	Retest: "Rollback recommendation or investigate further".	**PASS**

5. Final Validated Capabilities

- Significant improvement detection using a configurable threshold.
- High/significant/insignificant improvement classification.
- Plan-change evidence including Seq Scan to Index Scan transitions.
- Index and composite-index recommendation validation.
- Successful / unsuccessful / not-applicable recommendation status.
- Workload-wide improvement detection across multiple queries.
- Separate planning-time and execution-time improvement reporting.
- Row-processing reduction as supporting evidence.
- Before/after reporting for plan, cost, rows, planning time, and execution time.
- Rollback or further-investigation guidance when a recommendation does not improve performance.

6. Final Conclusion

FINAL ACCEPTANCE: PASS - 25 / 25 test cases passed (100%).

Audit note: This report summarizes the final response evidence supplied during interactive Swagger testing. It does not claim screenshot proof where only textual JSON payloads were supplied, and it does not assert independent SQLite execution outside those supplied responses.

QUERY PERFORMANCE PROJECT

Edge Case Validation
Complete Status Report

EC-01 to EC-30

Final Result	**30 PASS / 0 FAIL**
Pass Rate	**100%**
Coverage	EC-01 through EC-30
Report Basis	Completed edge-case test outcomes supplied for final documentation

FINAL STATUS: 30 / 30 PASS

All listed edge-case scenarios are documented as PASS against their expected behavior.

1. Executive Summary

This report consolidates the final status of the 30 edge cases defined for the Query Performance Project. The covered scenarios include missing measurements, zero and extreme timings, significance thresholds, regressions, plan/runtime disagreement, repeated-measurement stability, cache and workload fairness, data/statistics changes, multi-change attribution, query identity, database/permission failures, index lifecycle, and insufficient-evidence handling. All 30 cases are recorded as PASS, giving a final pass rate of 100%.

2. Final Status Matrix

EC ID	Edge Case	Expected Behaviour	Status
EC-01	Before execution missing	Cannot calculate improvement	**PASS**
EC-02	After execution missing	Cannot validate optimization	**PASS**
EC-03	Before time = 0	Handle division safely	**PASS**
EC-04	Very small timing difference	Use configured significance threshold	**PASS**
EC-05	After time > before time	Report regression	**PASS**
EC-06	Same execution time	Report no significant change	**PASS**
EC-07	Plan changes but runtime doesn't improve	Don't claim success	**PASS**

EC ID	Edge Case	Expected Behaviour	Status
EC-08	Runtime improves but plan unchanged	Report runtime improvement	**PASS**
EC-09	One unusually fast/slow execution	Use repeated measurements where possible	**PASS**
EC-10	High execution-time variance	Report unstable measurement	**PASS**
EC-11	Cache warm vs cold	Ensure before/after comparison is fair	**PASS**
EC-12	Data changes between tests	Mark comparison as potentially affected	**PASS**
EC-13	Concurrent workload changes	Reduce confidence in comparison	**PASS**
EC-14	Multiple indexes added	Can't isolate individual index benefit	**PASS**
EC-15	Multiple tuning changes	Attribute improvement cautiously	**PASS**
EC-16	Query already fast	Avoid unnecessary optimization	**PASS**
EC-17	Improvement disappears later	Detect later regression	**PASS**
EC-18	Recommendation gives no benefit	Mark recommendation unsuccessful	**PASS**
EC-19	Improvement only for one parameter	Test representative parameters	**PASS**
EC-20	Different query between tests	Don't compare as same query	**PASS**
EC-21	Query plan unavailable	Use available metrics and lower confidence	**PASS**
EC-22	Database unavailable	Handle measurement failure	**PASS**
EC-23	Permission error	Report clearly	**PASS**
EC-24	Very large execution time	Handle numeric values safely	**PASS**
EC-25	Very small execution time	Avoid misleading percentage calculations	**PASS**
EC-26	Table size changed	Record data-size difference	**PASS**
EC-27	Statistics changed	Record ANALYZE/statistics event if available	**PASS**
EC-28	Index removed after testing	Record optimization no longer active	**PASS**
EC-29	Index exists but planner ignores it	Don't call optimization successful automatically	**PASS**
EC-30	Improvement cannot be confirmed	Report insufficient evidence	**PASS**

3. Detailed Validation Notes

The following entries document the scenario, expected behaviour, final recorded result, and the behaviour used to justify the PASS status.

EC-01 - Before execution missing

Status	**PASS**
Expected	Cannot calculate improvement
Observed / validated behaviour	Validation prevents improvement calculation when the baseline execution value is missing.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-02 - After execution missing

Status	**PASS**
Expected	Cannot validate optimization
Observed / validated behaviour	Validation prevents optimization success/failure classification when the after-execution value is missing.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-03 - Before time = 0

Status	**PASS**
Expected	Handle division safely
Observed / validated behaviour	Zero baseline timing is handled without divide-by-zero or misleading improvement percentages.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-04 - Very small timing difference

Status	**PASS**
Expected	Use configured significance threshold
Observed / validated behaviour	Minor timing differences are treated as insignificant unless they cross the configured threshold.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-05 - After time > before time

Status	**PASS**
Expected	Report regression
Observed / validated behaviour	A slower after-execution measurement is classified as a regression, not as an optimization success.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-06 - Same execution time

Status	**PASS**
Expected	Report no significant change
Observed / validated behaviour	Equal before/after execution times are classified as no significant change.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-07 - Plan changes but runtime doesn't improve

Status	**PASS**
Expected	Don't claim success
Observed / validated behaviour	A query-plan change alone is not treated as successful optimization without runtime benefit.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-08 - Runtime improves but plan unchanged

Status	**PASS**
Expected	Report runtime improvement
Observed / validated behaviour	Runtime improvement is reported even when the execution plan remains unchanged.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-09 - One unusually fast/slow execution

Status	**PASS**
Expected	Use repeated measurements where possible
Observed / validated behaviour	Repeated measurements are preferred so a single outlier does not drive the final conclusion.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-10 - High execution-time variance

Status	**PASS**
Expected	Report unstable measurement
Observed / validated behaviour	High timing variance is flagged as unstable rather than being presented as a reliable optimization result.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-11 - Cache warm vs cold

Status	**PASS**
Expected	Ensure before/after comparison is fair
Observed / validated behaviour	The comparison accounts for cache-state fairness so cold and warm runs are not treated as directly equivalent evidence.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-12 - Data changes between tests

Status	**PASS**
Expected	Mark comparison as potentially affected
Observed / validated behaviour	Data changes are recorded as a factor that may affect the validity of the before/after comparison.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-13 - Concurrent workload changes

Status	**PASS**
Expected	Reduce confidence in comparison
Observed / validated behaviour	Concurrent workload changes lower confidence in the optimization attribution.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-14 - Multiple indexes added

Status	**PASS**
Expected	Can't isolate individual index benefit
Observed / validated behaviour	When several indexes are introduced together, benefit is attributed cautiously and not assigned to one index without evidence.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-15 - Multiple tuning changes

Status	**PASS**
Expected	Attribute improvement cautiously
Observed / validated behaviour	Attribution is marked as combined and confidence is lowered when multiple tuning changes occur.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-16 - Query already fast

Status	**PASS**
Expected	Avoid unnecessary optimization
Observed / validated behaviour	Returns a not-needed/not-applicable outcome and avoids unnecessary optimization for an already-fast query.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-17 - Improvement disappears later

Status	**PASS**
Expected	Detect later regression
Observed / validated behaviour	Later validation detects regression when an earlier performance improvement does not persist.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-18 - Recommendation gives no benefit

Status	**PASS**
Expected	Mark recommendation unsuccessful
Observed / validated behaviour	A recommendation with no measurable benefit is marked unsuccessful.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-19 - Improvement only for one parameter

Status	**PASS**
Expected	Test representative parameters
Observed / validated behaviour	Representative parameter measurements are checked and parameter-specific gains are not generalized.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-20 - Different query between tests

Status	**PASS**
Expected	Don't compare as same query
Observed / validated behaviour	Different query identities are rejected from direct before/after comparison.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-21 - Query plan unavailable

Status	**PASS**
Expected	Use available metrics and lower confidence
Observed / validated behaviour	Execution metrics are still used, but confidence is reduced when the plan is unavailable.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-22 - Database unavailable

Status	**PASS**
Expected	Handle measurement failure
Observed / validated behaviour	Database unavailability is returned as a measurement failure with a clear failure classification.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-23 - Permission error

Status	**PASS**
Expected	Report clearly
Observed / validated behaviour	Permission failures are reported explicitly rather than being misclassified as performance results.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-24 - Very large execution time

Status	**PASS**
Expected	Handle numeric values safely
Observed / validated behaviour	Large finite timing values are handled safely without overflow or invalid arithmetic.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-25 - Very small execution time

Status	**PASS**
Expected	Avoid misleading percentage calculations
Observed / validated behaviour	Very small timings avoid misleading percentage calculations and unnecessary optimization claims.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-26 - Table size changed

Status	**PASS**
Expected	Record data-size difference
Observed / validated behaviour	Before/after table-size or row-count differences are recorded and confidence is adjusted.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-27 - Statistics changed

Status	**PASS**
Expected	Record ANALYZE/statistics event if available
Observed / validated behaviour	Statistics refresh/ANALYZE events are recorded as potential contributors to performance change.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-28 - Index removed after testing

Status	**PASS**
Expected	Record optimization no longer active
Observed / validated behaviour	The system records that the optimization is inactive when an index used during testing is later removed.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-29 - Index exists but planner ignores it

Status	**PASS**
Expected	Don't call optimization successful automatically
Observed / validated behaviour	Index existence alone is insufficient; planner usage and measured benefit are required before claiming success.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

EC-30 - Improvement cannot be confirmed

Status	**PASS**
Expected	Report insufficient evidence
Observed / validated behaviour	When evidence is incomplete or conflicting, the result is insufficient evidence rather than a false success.

Assessment: Observed behaviour matches the defined expected behaviour; final status is PASS.

4. Final Acceptance Summary

- Missing before/after measurements are handled without false optimization claims.
- Zero, very small, very large, equal, improved, and regressed timings are classified safely.

• Plan changes and runtime changes are evaluated independently so plan-only changes do not create false success.

• Repeated measurements, variance, cache state, data changes, and concurrent workload changes are treated as evidence-quality factors.

• Multiple indexes or tuning changes are attributed cautiously instead of claiming a single cause without proof.

• Query identity, plan availability, database availability, and permission failures are handled explicitly.

• Index removal, planner non-use, table-size changes, and statistics events are recorded when relevant.

• When improvement cannot be confirmed, the system reports insufficient evidence instead of overstating success.

FINAL ACCEPTANCE: PASS - 30 / 30 EDGE CASES PASSED (100%)

Audit note: This document is a consolidated status report of the completed test outcomes. It does not claim additional independent execution beyond the recorded test results used to prepare the report.